

First degree.

Consider an equation $Y' + a(t) \cdot Y = 0$. Let's solve this.

If $Y(t_0) = 0$ for some $t_0 \in I$, then since $\tilde{Y} \equiv 0$ is a solution s.t. $\tilde{Y}(t_0) = 0$, the Uniqueness theorem gives $Y \equiv 0$. Now, assume $Y > 0$ for all $t \in I$.

Then, we obtain $\frac{Y'}{Y} = (\log Y)' = -a(x)$, so

$$\log Y = - \int_{t_0}^t a(x) dx \Rightarrow Y = \exp\left(- \int_{t_0}^t a(x) dx\right)$$

Non-Homogeneous Case

Consider an equation $Y' + a(t) \cdot Y = b(t)$.

We already know the Homogeneous Solution

$$Y_H = \exp\left(- \int_{t_0}^t a(x) dx\right)$$

And, the Particular Solution is

$$Y_p(t) = Y_H(t) \cdot \int_{t_0}^t \frac{b(x)}{Y_H(x)} dx$$

Therefore, the general Solution is

$$Y(t) = Y_H(t) \cdot \left[C + \int_{t_0}^t \frac{b(x)}{Y_H(x)} dx \right]$$

